

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First / Second Semester of B. Tech. Examination (All branch)

MAY 2014

EE 102 Fundamentals of Electrical Engineering

Date: 22.05.2014, Thursday Time: 01:30 p.m. To 04:30 p.m.

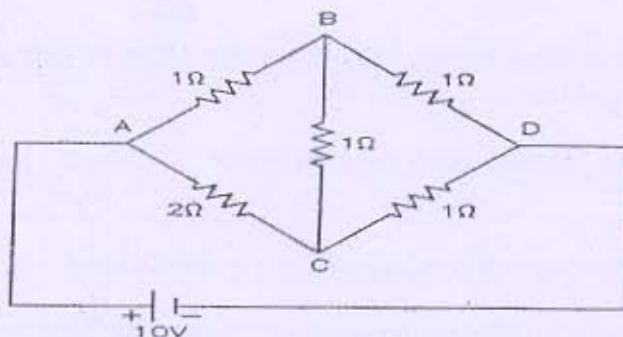
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 (a) Explain the factors affecting the resistance of a conducting material. [4]
- (b) State and explain Faraday's laws of electromagnetic induction. [3]
- Q – 2 (a) Using Star-Delta Transformation only find out Current drawn from the source in given circuit. [7]



- (b) Derive the formula for star to delta and delta to star transformation with necessary diagram. [7]

OR

- Q – 2 (a) A parallel plate capacitor with five plates has each plate one meter square area at a distance of 1.5 mm and a dielectric of relative permittivity 3.5. If the capacitor is charged to 1000 volt, how much energy will be stored in it. [7]
- (b) Explain charging and discharging of a capacitor [7]
- Q – 3 (a) Give the Comparison of Electric circuit and magnetic circuit. [7]
- (b) A ring has a mean diameter of 21 cm and a cross sectional area of 10 Cm². The ring is made up of semi circular sections of cast iron and cast steel, with joint having reluctance equal to an air gap of 0.2 mm. Find The amp-turns required to produce a flux of 8×10^{-4} wb. The relative permeability of cast steel and cast iron are 800 and 166 respectively. [7]

Neglect fringing and leakage effect.

OR

- Q – 3 (a) State and explain Ohm's law and its limitations. [7]
- (b) State and explain Kirchhoff's laws with necessary diagram. [7]

SECTION – II

- Q – 4 Define the following terms: [7]
- (1) Magnetic flux density
 - (2) Magnetic field intensity
 - (3) MMF
 - (4) Reluctance
 - (5) Permeance
- Q – 5 (a) Derive expressions for impedance, power factor and current for an R - L series circuit when an a.c. voltage is applied to it. Draw also the phasor diagram. [7]
- (b) Explain the term power factor in connection with ac circuits. Draw impedance triangle for R - L , R - C , & R - L - C series circuit. [7]

OR

- Q – 5 (a) Define average value. Explain average value in case of sinusoidal current & half wave rectifier. [7]
- (b) Explain the form factor and peak factor [7]
- Q – 6 (a) An alternating emf is represented by $e = 400 \sin 314 t$ [7]
- Determine : (i) Maximum value (ii) Frequency (iii) Time period (iv) Angular frequency
- (b) Define capacitance. Derive an expression for the capacitance of a parallel plate capacitor. [7]

OR

- Q – 6 (a) Derive the relationship between line voltage & phase voltage in a 3-phase star-connected circuit. [7]
- (b) Derive the relationship between line current & phase current in a 3-phase delta-connected circuit. [7]
